ARAS OZDEMIR

Quality • Manufacturing • Process & Test Engineering

ao473@cornell.edu | (607) 262-5047 | Ithaca, NY 14850 | linkedin.com/in/aras-ozdemir

SUMMARY

Mechanical and biomedical engineer with hands-on quality, design, and test experience across FDA-regulated medical device manufacturing. Lean Six Sigma Black Belt who pairs product and process design with 510(k), CAPA, and IQ/OQ/PQ validation work, and a record of building tooling and workflows that cut hours and cost out of validation.

EXPERIENCE

Quality Engineer, Fresenius Kabi Nov 2024 – Aug 2025

Medical Device R&D / Regulatory *Frankfurt, Germany*

- Supported preparation of Class II 510(k) submission documentation, contributing directly to regulatory filing.
- Investigated device complaints and nonconformances using root cause analysis (5-Why, Ishikawa) to drive **corrective and preventive actions (CAPA)** in an R&D environment.
- Revised **40+ technical drawings** for FDA 21 CFR Part 820 and ASME Y14.5 compliance, standardizing GD&T tolerances.
- Engineered an automated magnetic hatch durability tester (10,000 cycles), **cutting ~28 manual labor hours per validation run** and improving repeatability by minimizing operator variability.
- Reengineered validation workflows to remove steps and enable material reuse, **improving efficiency ~20% and reducing cost 80%.**

Design Engineer, B. Braun Medical Jan 2024 – May 2024

Catheter Connector Development *State College, PA*

- **Improved catheter connector grip strength by 48%** through iterative CAD prototyping and COMSOL-based FEA while maintaining customer-specified flow-rate requirements.
- Analyzed failure modes of secondary connector unclamping during procedures through mechanical testing.
- Conducted validation testing and ensured documentation complied with FDA design-control standards.

Research Assistant, Pennsylvania State University Jan 2023 – Jan 2024

Antimicrobial Agents Lab *State College, PA*

- Built a MATLAB app with automated signal filtering and data-validation checks, reducing manual analysis time and improving measurement consistency.
- Authored and maintained SOPs for bacteriophage–bacteria interaction analysis, ensuring reproducibility and data integrity across multiple test cycles.

PROJECTS

Engineering Lead — ASML Reticle-Handling System, Cornell Aug 2025 – May 2026

- Led an ASML-sponsored capstone integrating a robotic handling arm for reticle transport into Cornell's Mechatronics Studio.
- Conducted FMEA-based risk assessments to identify operational hazards and define mitigation strategies.
- Designed a sensor-integrated fail-safe shutdown system that disables the robot under unsafe trajectory conditions.

Lean Six Sigma Black Belt — Disc Flight Optimization (DMAIC), Cornell Sep 2025 – Dec 2025

- Ran a full DMAIC study with SIPOC diagrams, process maps, and a VOC framework to define CTQ metrics.
- Designed a factorial DOE (200+ runs) with regression and Response Surface models in R, isolating statistically significant interactions (**$p < 0.001$**).
- Validated effectiveness with bootstrapping (1,000 iterations) and permutation tests, then built control plans to sustain gains.

EDUCATION

Cornell University Ithaca, NY

M.Eng., Mechanical Engineering: GPA 3.95 *May 2026*

Pennsylvania State University State College, PA

B.S. Biomedical Engineering & B.S. Mechanical Engineering: GPA 3.55 *May 2024*

SKILLS & CERTIFICATIONS

Quality Systems: CAPA, FMEA / DFMEA, DHF, Design Control, NCR, IQ/OQ/PQ Validation, ISO 13485, ISO 9001, 510(k), Matrix Requirements (QMS)

Technical: SolidWorks, ANSYS, COMSOL Multiphysics, MATLAB, Python, R, Excel, LaTeX, GD&T

Certifications & Awards: Lean Six Sigma Black Belt (LSSBB); The German Scholarship

Languages: English (Fluent), German (Fluent), Turkish (Intermediate), Spanish (Beginner)